

$$f(w) = (w - z_1) \dots (w - z_n)$$

$$|f(z_j)| \in \underbrace{\{ |f(w)| : |w| = 1 \}}_{\rightarrow A} \quad [\because |z_j| = 1]$$

$$\Rightarrow 0 \in \{ |f(w)| : |w| = 1 \}$$

Now, f is entire fun $\Rightarrow f$ is holomorphic on $\{ |w| < 1 \}$
 and continuous on $\overline{\{ |w| < 1 \}} = \{ |w| \leq 1 \}$. Then by
 Maximum Modulus principle we get, $\exists w_0, \text{ s.t. } |w_0| = 1$

$$|f(0)| \leq |f(w_0)| \quad \left[\text{as maximum value of } |f| \text{ over the domain } \{ |w| \leq 1 \} \text{ is attained at its boundary.} \right]$$

$$\Rightarrow 1 \leq |f(w_0)| \quad \& \quad M = |f(w_0)| \in A$$

Now, $\{ |w| = 1 \}$ is connected & compact subset of $\mathbb{C}$.

$$|f| \text{ is cont fun on } \mathbb{C} \Rightarrow |f|(\{ |w| = 1 \}) = A$$

$\Rightarrow A$ is connected & cpt.
 subset of $\mathbb{R}$

$\Rightarrow A$ is closed bdd & (interval or singleton)

But clearly A is not singleton & $0 \& M \in A$

$\Rightarrow A$ is a closed bdd interval $\Rightarrow [0, M] \subseteq A$

$$\text{as } M \geq 1 \Rightarrow [a] \subseteq [aM] \subseteq A [mod M]$$